

AZGAR. Barakathulla

barkathmom@gmail.com

Phone:9110691171

Summary:

- Innovative Tool Developer and AUTOSAR Software Integration Specialist with 6+ years of experience in automotive and EV domain (Tier-1 & OEM, 2W/4W platforms)
- Skilled in software development, automation, and toolchain optimization, delivering scalable solutions that improve efficiency and user experience
- Strong expertise in AUTOSAR Classic architecture, including BSW, communication stack, diagnostics (UDS), OS, and RTE
- Hands-on experience with Vector tools: DaVinci Configurator, CANoe, and CANalyzer for ECU configuration and network validation
- Proficient in version control and CI/CD tools such as Git, Jenkins, and Jira for efficient software lifecycle management
- Experienced in debugging integration failures and resolving system-level issues, ensuring stable ECU software releases
- Extensive experience in HiL test bench development using dSPACE ControlDesk, including system validation and fault analysis
- Strong background in MATLAB/Simulink model development, maintenance, and validation for embedded control systems
- Expertise in root cause analysis (RCA) with implementation of quick bug fixes and long-term corrective solutions
- Proven track record in simulation workflow optimization, achieving ~30% efficiency improvement
- Developed automation scripts and tools, saving approximately ~20 hours per month by reducing manual effort
- Built diagnostic and analytical tools (PythonBAM, Confluence dashboards) for improved monitoring and debugging
- Experience across complete Software Development Life Cycle (SDLC) including development, integration, testing, and validation
- Strong knowledge of ISO 26262 functional safety, applied in systems like BMS and ACU
- Recognized for structured problem-solving, creativity, and clear technical communication in cross-functional environments

Professional Experience:

Akkodis Technologies, Bangalore (Client: Stellantis) Sr. Software Engineer – Tool Developer (Simulation & Automation) November 2024 – Present

Responsibilities & Achievements:

- Developed power electronics plant models using MATLAB and Simulink and integrated them with HiL platforms such as dSPACE for inverter and control software validation.
- Developed **MATLAB/Simulink plant models** integrated with **dSPACE HiL systems**
- Designed and executed **HiL test plans with requirement traceability**
- Developed **ControlDesk dashboards** for real-time monitoring
- Performed **fault analysis and root cause investigation (RCA)**
- Optimized **simulation workflows (~30% efficiency improvement)**
- Automated workflows using **Python/VBA (~20 hours/month savings)**
- Architected and maintained **HiL systems for ECU validation**
- **Advanced / Real-Time Contributions**

- Developed and enhanced **ECU test tools** for validation and debugging
- Worked on **ETM cluster functionalities** and feature validation
- Validated **vehicle-level functionalities and ECU interactions**
- Performed **basic diagnostics (UDS), ECU flashing, and fault analysis**
- Validated **CAN/LIN communication using CANoe, CANalyzer**
- Debugged system-level issues using **Trace32**
- Ensured compliance with **ASPICE and ISO 26262 standards**

Technical Skills

- AUTOSAR (BSW, COM, DCM, OS, NvM), ECU Integration
- MATLAB, Simulink, dSPACE HiL (ControlDesk)
- CAN, LIN, UDS Diagnostics, ECU Flashing
- Vector Tools: CANoe, CANalyzer, CANape
- Debugging: Trace32 (Lauterbach)
- Programming: Embedded C, C++, Python, VBA
- ECU Test Tools, ETM Cluster, Vehicle Functionalities
- Git, CI/CD, CodeBeamer, DOORS
- Standards: ISO 26262, ASPICE, Basic Cybersecurity

Work Description:

- Working as Sr. Software Engineer Akkodis Technologies Bangalore With Client Stellantis Tool Developer Simulation & Automation Nov 2024 to Till Date.
- Worked as Software Engineer in APTIV Bangalore from Nov 2019 to December 2022
- Worked as Software Engineer in Wipro limited, Bangalore from Feb 2023- to May 2023 Client, LG-Soft.
- Worked as SR-Software Engineer in ALTEN GLOBAL TECHNOLOGY SOLUTIONS PRIVATE LIMITED) Bangalore from JUNE-2023 to MARCH-2024

Professional Experience:

ALTEN Global Technology Solutions Private Limited, Bangalore | June 2023 – March 2024

Project Summary:

- Developed the ABS Control system with integrated CAN Communication and extensive BMS Implementation to ensure safe vehicle dynamics and optimal battery performance.
- BMS, Modeling, and Algorithm Development
- Developed and refined MATLAB algorithms to continuously calculate the maximum safe charge current limit ($I_{\text{charge_max}}$) based on cell temperature, voltage, and SoC. This value defined the State of Function (SoF) communicated to the motor controller and ABS system.
- Determined the maximum regenerative current the battery could safely accept (SoF) based on cell temperature and SoC .
- Implemented core BMS models including CAN model, Fault model, State machine, Contactor control, Contactor diagnostic, and measurement models.
- Ensured that during active ABS cycling, the regenerative current was safely managed or paused to maintain vehicle stability.
- Generated State flow and Simulink models as per requirements, followed by Unit Testing of Model,

- Auto code generation, and checking adherence to \text{MAAB} guidelines.
- Performed Data analysis, \text{MIL} (Model-in-the-Loop), and \text{SIL} (Software-in-model).

Technical Skills:

MATLAB, Simulink, Stateflow, Embedded C++ (C++11/14),
Model-Based Development (MBD), BMS Algorithm Development,
CAN Communication, Vector CANoe, CANalyzer, MIL/SIL Testing, Simulink Coder,
Embedded Coder, MAAB Modeling Guidelines, Automotive Control System

Wipro Limited, Bangalore (Client: LG-Soft) | February 2023 – May 2023

Project Summary:

- Developer role for the LG In-Vehicle Infotainment project.
- Responsibilities included analyzing software requirements, ensuring bidirectional traceability, and setting up/maintaining the software test environment.
- Developed and executed manual and/or automated tests on virtual environments, target ECU, and HIL. software architecture and the software components as well as their interfaces

APTIV, Bangalore | November 2019 – December 2022

Project Summary:

- Experience in the Airbags Development (Passive Safety) project, involving the development and configuration of the ACU (Airbag Control Unit).
- Implemented RCAN communication between different ECUs and configurations.
- Involved in design, development, and testing, and deriving software requirements for the COM module and DCM. (Bsw Com Stack had Worked the Os Configuration in Davinci Tool Layer for Task to Events, Alarms and Scheduling the behave in an ECUs and configured/implemented UDS Applications services for DCM.
- Handled BMS with body control and thermal management, and fail-safe operations in applications.

Key Skills:

Technical Skills

Embedded Software Development

- Embedded C, C++ (C++11 / C++14), Python
- Firmware development for automotive ECUs
- Bootloader development and ECU flashing

Model-Based Development (MBD)

- MATLAB
- Simulink
- Stateflow
- Algorithm development and model validation
- Auto code generation using **Embedded Coder / RTWEC**

Automotive Communication Protocols

- CAN, CAN FD
- SPI
- I2C
- LIN, FlexRay, Ethernet (Basic knowledge)

HIL & Software Testing

- Hardware-in-the-Loop (HIL) Testing
- Model-in-the-Loop (MIL) Testing
- Software-in-the-Loop (SIL) Testing
- Unit Testing

- Software Integration Testing

ECU Configuration Tools

- DaVinci Configurator
- EB Tresos

ECU Programming & Flashing

- Vector vFlash (CAN, CAN FD, FlexRay, LIN, Ethernet)

Debugging Tools

- Trace32 (Lauterbach)
- Embedded system debugging and trace analysis

Requirement & ALM Tools

- IBM DOORS – Requirements management
- JIRA – Issue tracking and project management

Version Control & Collaboration

- Git
- SVN
- Bitbucket
- PTC Integrity / Codebeamer
- Confluence

Testing & Analysis Tools

- LMS Test Lab
- Data analysis and validation

Operating Systems

- Linux
- Windows

Process & Standards

- ASPICE Process Knowledge
- MAAB Modeling Guidelines

Education Qualification:

Jawaharlal Nehru Technological University, Bachelor of Technology in computer science and engineering 2017 Completed.

LANGUAGES: English, Hindi, Kannada